“Man invented language to satisfy his deep need to complain”

— Lily Tomlin

NLP1

Language

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https://nlp-illc.github.io/nlp1/2026/

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Previously: What makes NLP so difficult anyway?

Empiricists (like Locke and Hume):

> *"Nothing is in the intellect that was not first in the senses."*
>
> *— (often attributed to) Thomas Aquinas*

To which rationalists retorted:

> *"...except the intellect itself."*
>
> *— Leibniz*

You and I have a language faculty, once equipped to acquire language (when we were children), now fully equipped to use it. **Computers do not**.

Outline

1. What Language Is

2. How do we do it?

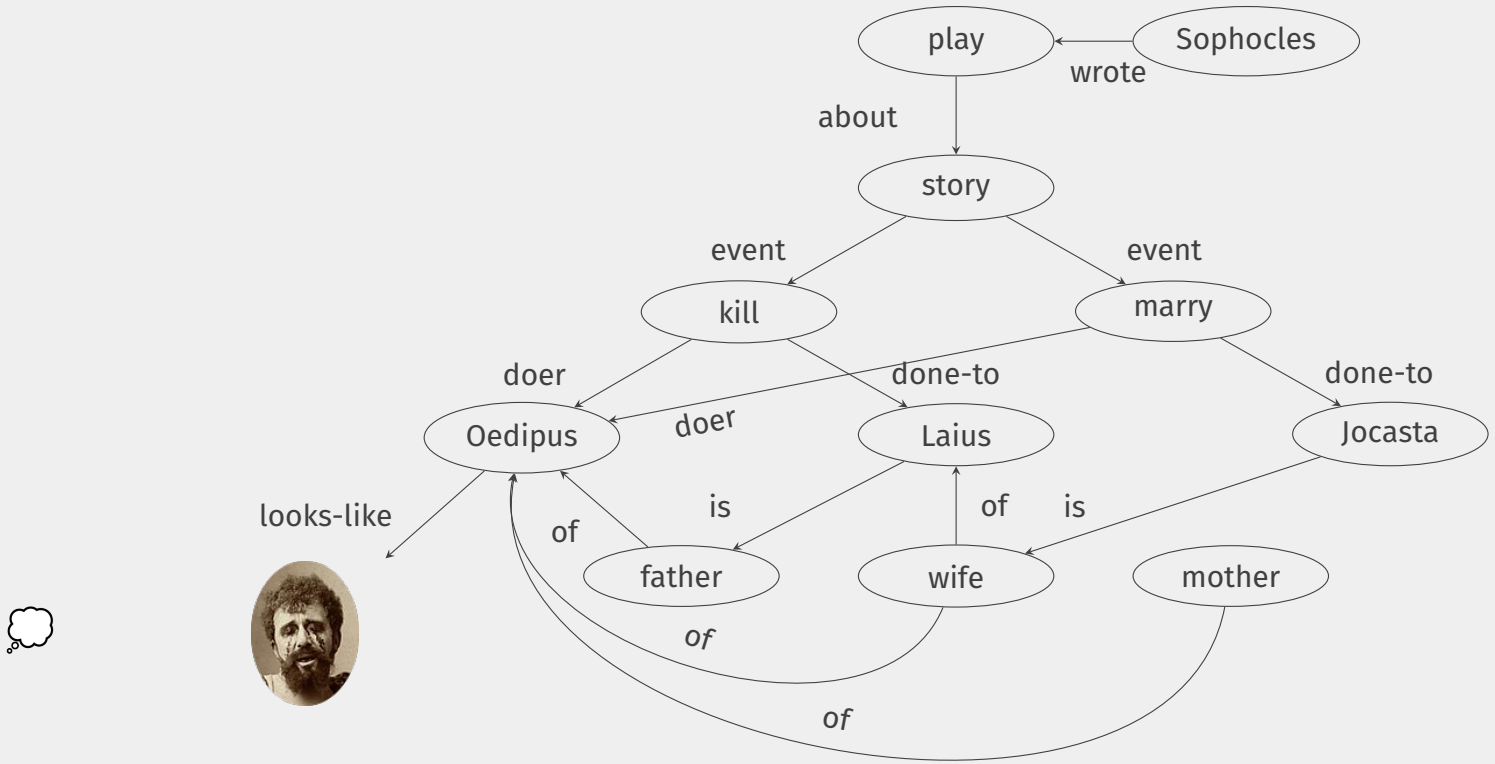
What Language Is

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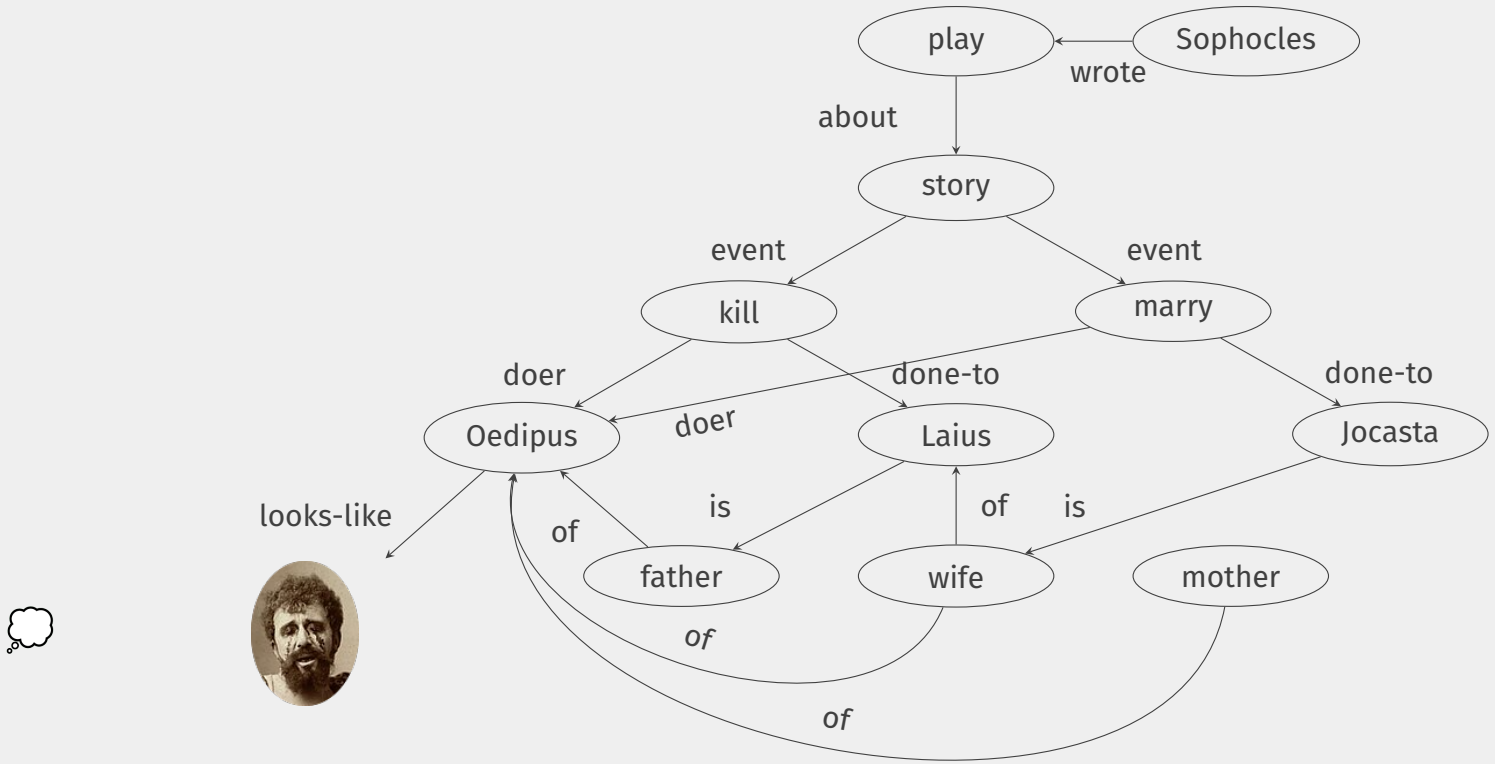
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You and I have a remarkable ability: by making noises with our mouths, we can cause each other to think precise thoughts.

Language is

- a complex skill that develops in the child spontaneously;
- deployed without awareness of its underlying logic;
- qualitatively the same in every individual;
- a distinct piece of the biological makeup of our brains;
- distinct from general abilities to process information or behave intelligently.

For such reasons, Language has been described as: a psychological faculty, a mental organ, a neural system, and a computational module.

An Instinct to Acquire an Art

In 1871, Darwin first articulated the conception of language as a kind of instinct

> *... man has an instinctive tendency to speak, as we see in the babble of our children; while no child has an instinctive tendency to brew, bake, or write ... man has an instinctive tendency to acquire an art*

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But the most famous argument that language is like an instinct comes from Noam Chomsky [6]:

> *The language each person acquires is a rich and complex construction* ***hopelessly underdetermined by the fragmentary evidence available to the child****. Nevertheless individuals in a speech community have developed essentially the same language. This fact can be explained only on the assumption that these individuals employ highly restrictive principles that guide the construction of grammar.*

Grammar in this passage is not the pedagogical guide to writing etiquette we are taught at school, but Language as a psychological faculty.

An Instinct to Acquire an Art

For a more modern, accessible and comprehensive argument, I recommend Pinker's *The Language Instinct* [12]:[2]

> *[instinct] conveys the idea that people know how to talk in more or less the sense that spiders know how to spin webs … spiders spin webs because they have spider brains, which give them the urge to spin and the competence to succeed.*

For a textbook account, I recommend Jackendoff [9].

[2] This class borrows heavily from it.

What is language for?

On this view, Language is

- our species' solution to the problem of transferring ideas from one mind to another;
- designed to solve problems that our ancestors met in the cognitive niche that their large, social brains inaugurated.

A common language connects members of a speech community into an information-sharing network.

What is the Evidence?

1. The universality of complex languages
2. Children reinvent language generation after generation
3. Language has its own seat (or seats) in the brain

The universality of complex languages

Cultural inventions vary widely in sophistication (some count with notches in ropes, some use computers), languages do not.[3]

Every language documented exhibits all of the resources necessary to consistently convey *who did what to whom*, *abstract concepts* and *complex reasoning*. For example, that's true of

- every 'non-standard dialect';
- the languages of societies isolated for tens of thousands of years;
- the sign languages of the deaf.

[3] [3, 10, 11, 13]

Children acquire language by reinventing it

Language acquisition by children does not amount to imitation:[4]

- they do not require instruction;
- they exhibit knowledge they should not have given their experience;
- they introduce complexity where none existed both in spoken and sign language.

[4] [4, 2, 7, 10]

Language has its own seat (or seats) in the brain

There are language-specific circuits in the brain:[5]

- Brain trauma in adulthood can spare (non-verbal) cognition but impair language (e.g., Broca's and Wernicke's aphasia);
- Specific language impairments may impede children from developing language on schedule, while having no impact on overall non-verbal cognition;
- Various forms of cognitive impairment (incl. with genetic basis such as Alzheimer's, William's syndrome) spare language.

[5] [8, 1]

What is this insight good for?

In the social sciences:

- To dispel ideas such that languages come in a hierarchy (with ‘proper languages’ at the top), that language shapes thought and that linguistic sophistication underlies cognitive virtuosity.
- To consolidate sign languages are true languages (they tap on the same psychological faculty as spoken languages).
- The list goes on...

For us, to set a realistic expectation about the challenge we face in getting computers to understand and generate language.

Other Hypotheses

Broadly speaking: language is a product of cultural evolution

- Cultural invention (Sapir): rejects innateness, attributes universality to universal communication needs, human physiology, and capacity for symbolic thought;
- Co-evolution (Tomasello): emphasises ontogeny (the developmental history of the organism) and frames language as a byproduct of the coevolution of human cognition and culture (with one driving the other in a Baldwin effect).

Summary

Can we then say that

> *"Man invented language to satisfy his deep need to complain"*
>
> — *Lily Tomlin*

The debate rages on

- but the exclusion of innate machinery is mostly untenable
- and, similarly, the importance of the co-evolution of brains and the cognitive niche that their holders lived in cannot be ignored.

Luckily (?) for us, we can leave the debate at that.

How do we do it?

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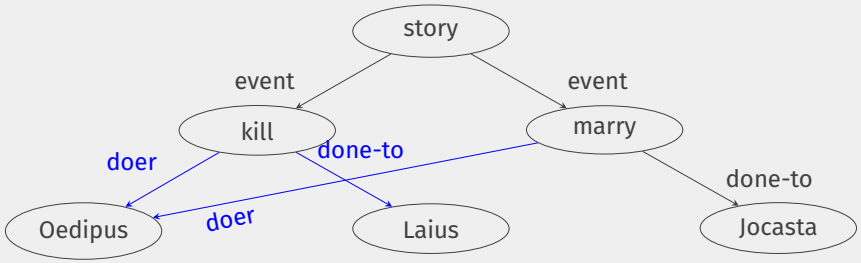
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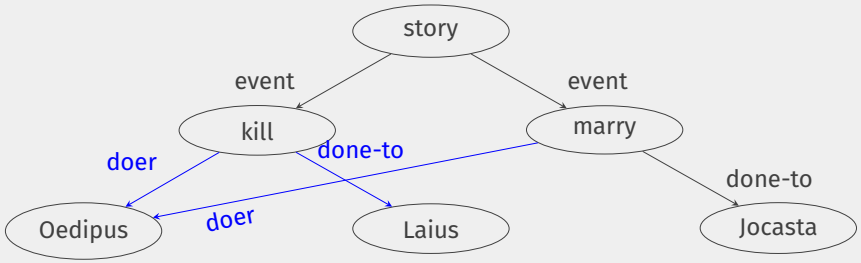


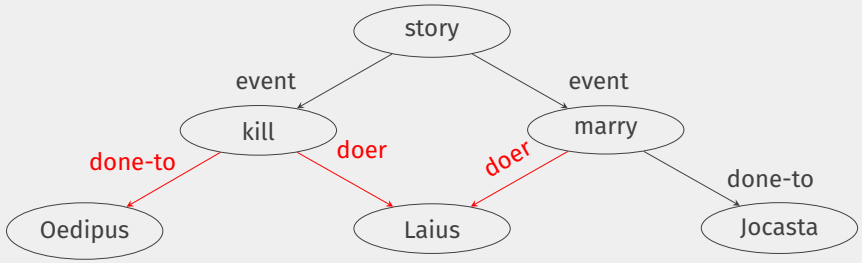
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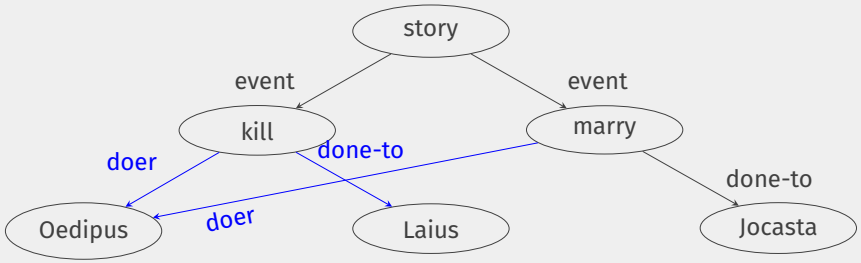


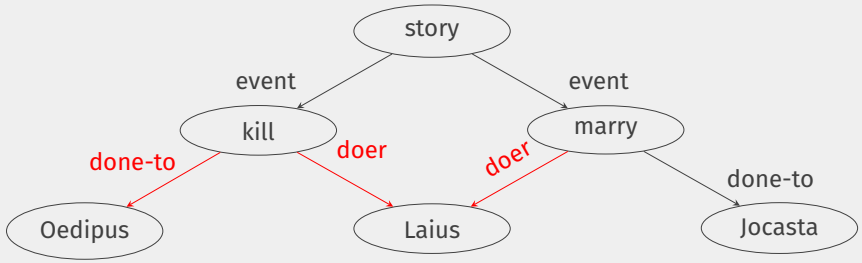
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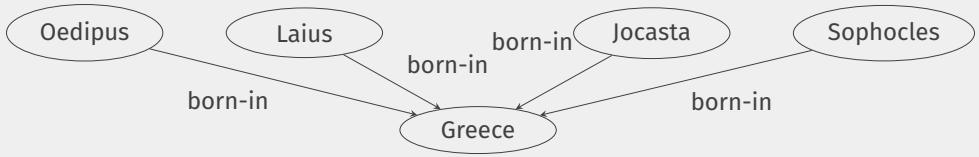
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1. killed Laius and married Jocasta
2. rather than was killed by Laius who married Jocasta
3. and not simply that the speaker is naming a bunch of Greeks.







How do we do it?

We know it because you and I, and every English speaker, have undergone an identical act of rote learning that links the sound of a word to its meaning.

And, we know a code to translate between orders of words and combinations of thoughts.

The arbitrariness of the sign

The pairing of a sound with a meaning is wholly conventional

- we memorise the correspondence in childhood using something like a mental dictionary;
- apprehending meaning from form is rare and mostly unreliable.

By fetching meaning from memory, using sound as a key, we are able to convey a concept from mind to mind virtually instantaneously.

These memorised items are what we call **listemes**.

Listemes are not the whole story

Test with words out of children's experience show that

- a 2 yo can form the plural of a new word;
- a 3 yo can inflect a new word for tense (and other grammatical paraphernalia)

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Derivation: English speakers can use word order to associate entirely different meanings with the same stretch of sound.

*The **chair** of the session moved the **chair** across the room.*

Word Machines

If all at our disposal were a large table in the brain, we would quickly run out of brain.

We must be equipped with a system of word formation that recombines memorised information—call it **morphology**.

Some of the evidence, again, comes from children. Across the world's languages (incl. sign languages), children

- make equivalent generalisations (and over-generalisations) on schedule;

Sentence Machines

If interrupted at random while producing a sentence, an English speaker can continue with, on average, any of 10 words.

Assume an adult is able to produce and comprehend sentences with 20 words. That brings us to 10^{20} sentences.

With no more than rote learning at our disposal:

- at 5 seconds per sentence, we would need a childhood of about a hundred trillion years
- one million times more brain

We use a code to translate between orders of words and combinations of thoughts—call it **grammar** (or **syntax**).

Infinite use of finite media

A grammar is a discrete combinatorial system in which a finite number of discrete elements (in this case words) are sampled, combined and permuted to create larger structures (in this case sentences).

The properties of the result are quite distinct from those of their elements, while still predictable.

In a discrete combinatorial system like language there can be an unlimited number of distinct combinations with an infinite range of properties.

Summary

The way language works then is that each person's brain contains

- a lexicon of symbols and the concepts they stand for—a mental dictionary,
- and a system that combines words to convey relationships amongst concepts—a mental grammar.

Next class we expand on these two modules:

- our goal is develop some appreciation for the complexity of the **data** we are about to embrace in this course.

Also, don't forget the action points (re groups and assessment).

References I



Ursula Bellugi, Amy Bihrle, and Sally Doherty.
Language, cognition, and brain organization in.
In *Developmental Behavioral Neuroscience: The Minnesota Symposia on Child Psychology, Volume 24*, page 201. Psychology Press, 2013.



Derek Bickerton.
***Roots of Language*.**
Karoma Publishers, Ann Arbor, MI, 1981.
Pp. xiii + 351.



Donald E Brown.
Human universals (1991): Reflections on its whence and whither.
Literary Universals Project, 2017.



Roger Brown.
***A First Language: The Early Stages*.**
Harvard University Press, Cambridge, MA, 1973.

References II


Noam Chomsky.
Aspects of the Theory of Syntax.
The MIT Press, Cambridge, 1965.


Noam Chomsky.
Language and Mind.
Harcourt Brace Jovanovich, New York, enlarged edition, 1972.


Stephen Crain and Mineharu Nakayama.
Structure dependence in grammar formation.
Language, 63(3):522–543, 1987.


Myrna Gopnik and Martha B. Crago.
Familial aggregation of a developmental language disorder.
Cognition, 39(1):1–50, apr 1991.


Ray Jackendoff.
Foundations of Language: Brain, Meaning, Grammar, Evolution.
Oxford University Press, 01 2002.

References III


Edward S Klima and Ursula Bellugi.
The signs of language.
Harvard University Press, 1979.


William Labov.
The logic of nonstandard english.
In *Language and poverty*, pages 153–189. Elsevier, 1970.


Steven Pinker.
The Language Instinct: How the Mind Creates Language.
William Morrow and Company, New York, NY, 1994.


Edward Sapir.
Language: An Introduction to the Study of Speech.
Harcourt, Brace & Company, New York, 1921.


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